

4. ABCD is a trapezium in which $AB \parallel DC$, BD is a diagonal and E is the mid – point of AD . A line is drawn through E parallel to AB intersecting BC at F

(see Fig.8.30).Show that F is the mid – point of BC.

GIVEN : * ABCD is a trapezium
 * BD is a diagonal
 * E is the mid – point of DA
 * $EG \parallel AB$ and $AB \parallel DC$

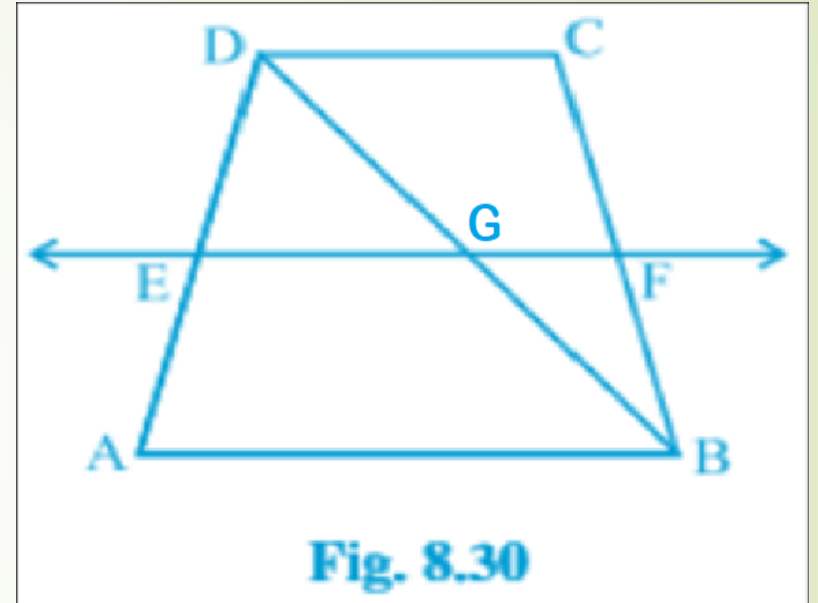
TO SHOW THAT : F is the mid – point of BC.

PROOF : In $\triangle ADB$, E is the mid – point of AD ,
 $EF \parallel AB$

$\Rightarrow$ G is the mid – point of BD (by converse of mid – point theorem)

$EF \parallel AB$, $AB \parallel CD$

$\Rightarrow$ $EF \parallel CD$

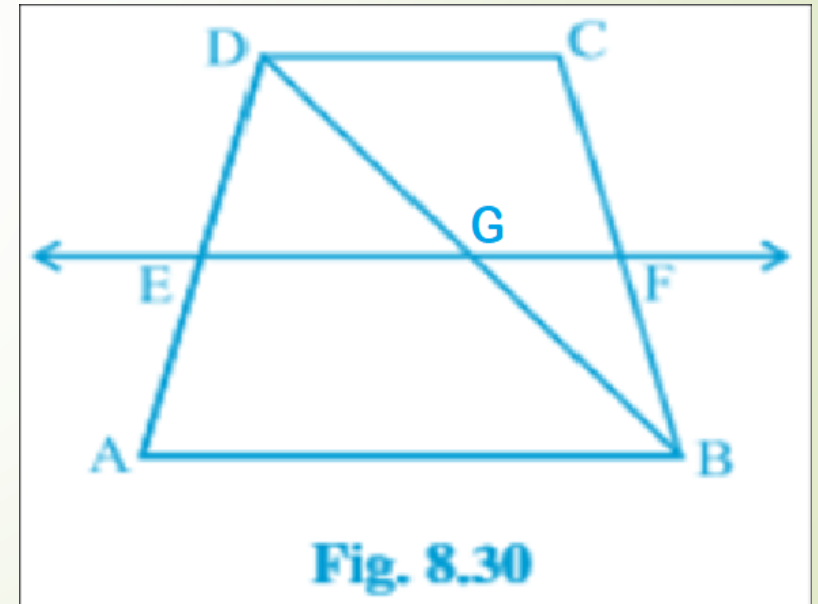


Let EF meet BD at G .

Hence in $\triangle BCD$, G is the mid – point of BD ,

$$GF \parallel CD$$

$\Rightarrow$ F is the mid – point of BC (Converse of mid – point theorem)



5. In a parallelogram ABCD , E and F are the mid – points of sides AB and CD respectively (see Fig.8.31) . Show that the line segment AF and EC trisect the diagonal BD .

GIVEN : * ABCD is a parallelogram

* E and F are the mid – points of sides AB and CD respectively

TO PROVE THAT :

Line segment AF and EC trisect the diagonal BD .

PROOF : ABCD is a parallelogram ,

$AB \parallel CD$, $AB = CD$

(i.e) $\frac{1}{2} AB \parallel \frac{1}{2} CD$ and $\frac{1}{2} AB = \frac{1}{2} CD$

$\Rightarrow AE \parallel CF$ and $AE = CF$

$\Rightarrow AECF$ is a parallelogram (by Theorem 8.8)

$\Rightarrow AF \parallel CE \rightarrow \textcircled{1}$

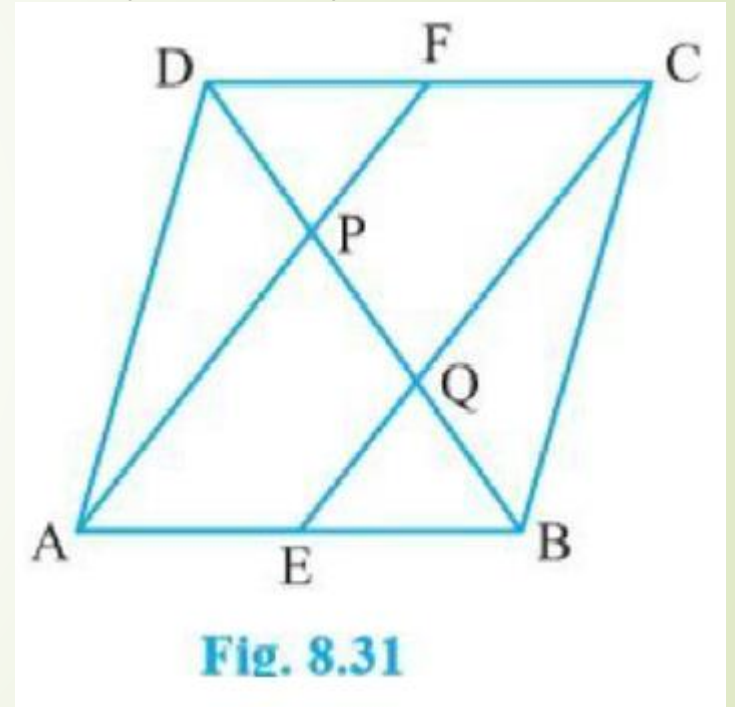


Fig. 8.31

In $\triangle BAP$, E is the mid - point of AB .

$\therefore EQ \parallel AP$ (from ①)

by converse of mid - point theorem

Q is the mid - point of BP

$\therefore BQ = PQ \rightarrow$ ②

In $\triangle DQC$, F is the mid - point of CD .

$\Rightarrow FP \parallel CQ$ (from ①)

$\Rightarrow P$ is the mid - point of DQ (by converse of mid - point theorem)

$\therefore DP = PQ \rightarrow$ ③

from ② and ③

$DP = PQ = BQ$

(i.e) BD is trisected .

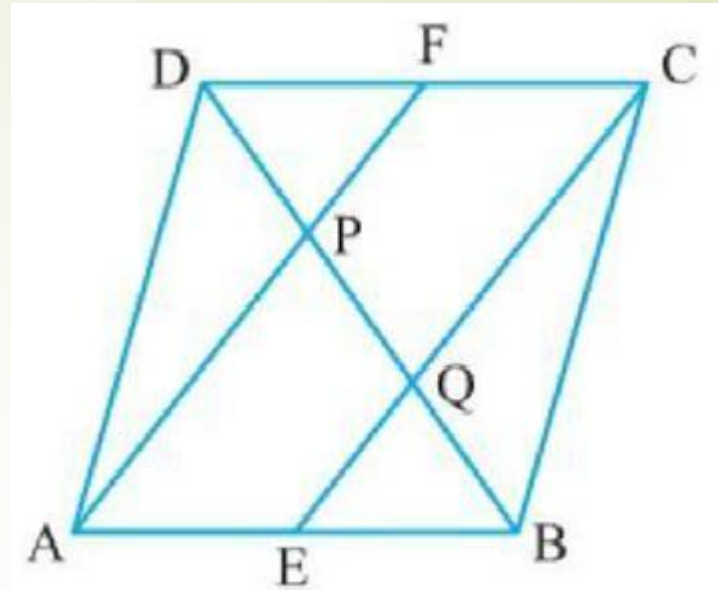


Fig. 8.31

6. Show that the line segment joining the mid – points of the opposite sides of a quadrilateral bisect each other.

GIVEN :

P , Q , R , S are mid – points of quadrilateral

TO PROVE THAT :

PR bisects SQ.

CONSTRUCTION :

Join PQ , QR , RS , SP and also join AC .

PROOF :

In $\triangle ADC$, S and R are mid – points of AD and CD.

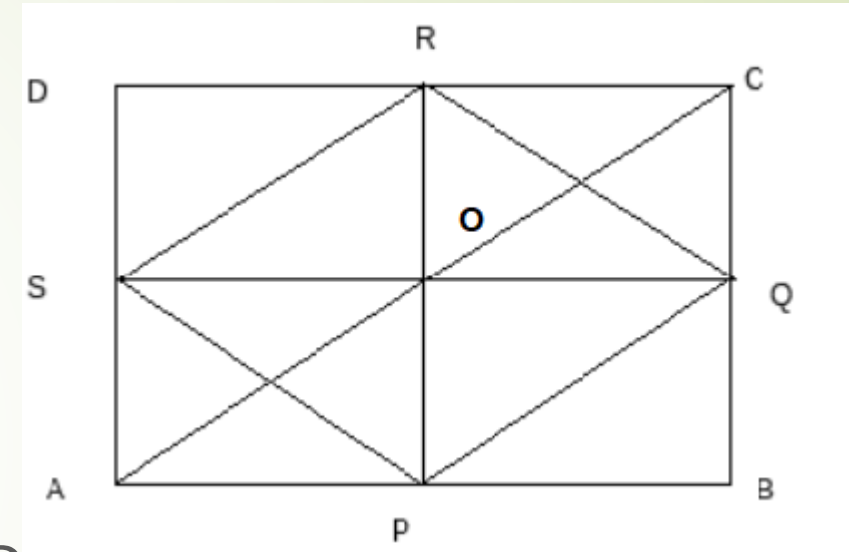
by mid – point theorem

$$SR \parallel AC , SR = \frac{1}{2} AC \rightarrow (1)$$

In $\triangle ABC$, P and Q are mid – points of AB and BC.

by mid – point theorem

$$PQ \parallel AC , PQ = \frac{1}{2} AC \rightarrow (2)$$

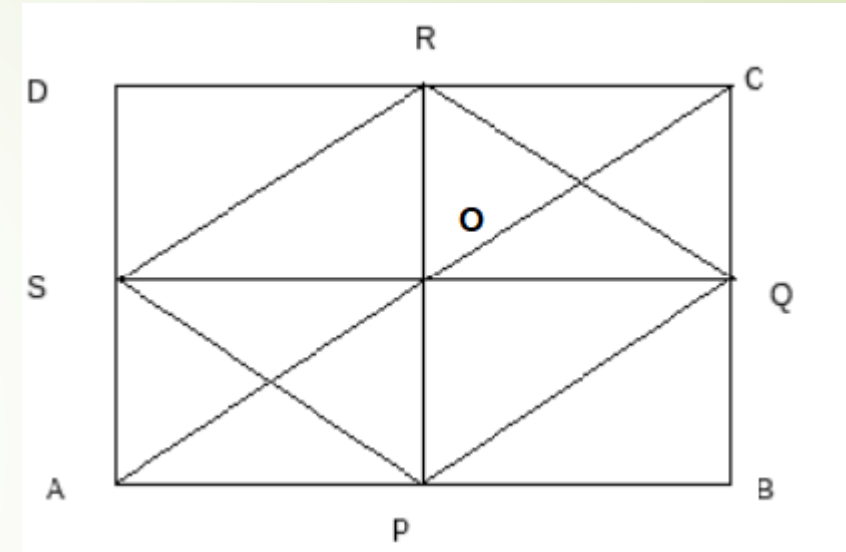


From ① and ②

$\Rightarrow PQ \parallel SR$, $PQ = SR$

$\therefore PQRS$ is a parallelogram.

$\Rightarrow$ Diagonals PR and SQ bisect each other.



7. ABC is a triangle right angled at C . A line through the mid – point M of hypotenuse AB and parallel to BC intersects AC at D . Show that

(i) D is the mid – point of AC .

(ii) $MD \perp AC$ (iii) $CM = MA = \frac{1}{2} AB$.

GIVEN : *ABC is a triangle with $\angle C = 90^\circ$

*M is the mid – point of AB , $MD \parallel BC$

TO PROVE THAT :

(i) D is the mid – point of AC .

(ii) $MD \perp AC$

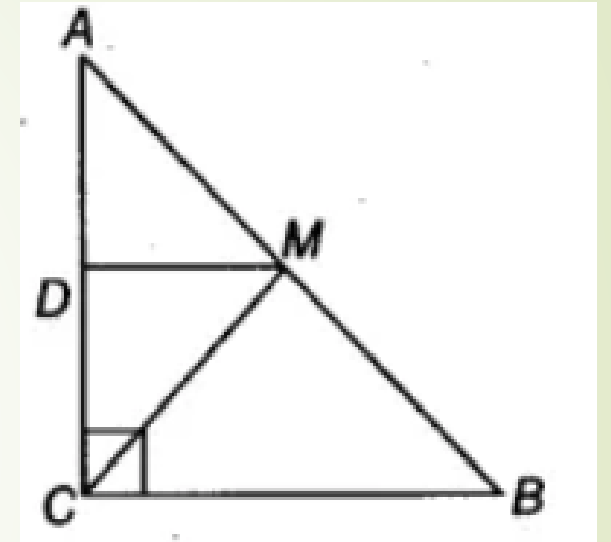
(iii) $CM = MA = \frac{1}{2} AB$.

PROOF :

(i) M is the mid – point of AB , $MD \parallel BC$

$\Rightarrow$ D is the mid – point of AC

(by converse of mid – point theorem)



(ii) $MD \parallel BC$ and AC is a transversal

$\therefore \angle ADM = \angle ACB = 90^\circ$ ($\because$ Corresponding angles)

$\Rightarrow MD \perp AC$

(iii) In $\triangle ADM$ and $\triangle CDM$

$AD = CD$

$\angle ADM = \angle CDM = 90^\circ$

$DM = DM$

$\triangle ADM \cong \triangle CDM$ (by SAS rule)

by cpct,

$AM = CM$

$\Rightarrow AM = CM = \frac{1}{2} AB$

